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优秀论文

论文题目

目标航发燃烧室的湍流燃烧数值模拟

作者简介



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摘要简介

AECSC (Aero engine combustion simulation code) 是北航仿真中心研发的针对航发燃烧室数值模拟的程序 (软件), 具有完全自主知识产权, 经过射流、钝体、预混、扩散、双旋流燃烧室等算例检验, 能实现复杂结构真实燃烧室的高保真模拟。此次优化采用负载均衡方法, 动态分配计算资源, 计算燃烧中的化学反应。对Flame D、E、F进行检验, 在鲲鹏计算平台与AMD服务器进行计算, 计算速度提高了近8%, 与实验结果对比, 计算准确性高。

Abstract

AECSC (Aero engine combustion simulation code) is developed by Beihang Aeroengine Simulation Research Center for numerical simulation of aero engine combustor, with completely independent intellectual property rights, through the jet, bluff body, premixed, diffusion, GTMC and other examples of test cases, to achieve the high-fidelity simulation of real combustors of complex structures. The optimization adopts the load balancing method to dynamically allocate computing resources to calculate the chemical reactions in combustion. Flame D, E, and F were examined, and the computation speed was increased by nearly 8% with Kunpeng computing platform and AMD servers, and the computation accuracy was high when compared with the experimental results.